Deterministic Falsification Tests and Algorithmic Bias Mitigation in the Upcoming Survey Era: A Dual-Scenario Approach for the Referential Relativity Theory (RRT)

Jean Coutinho Cortez
Independent Researcher

April 15, 2026

Abstract

Contemporary cosmology (ΛCDM) faces profound epistemological crises, increasingly relying on an unobservable "dark sector" to reconcile galactic dynamics with cosmic expansion. In this paper, I establish a strict, deterministic predictive agenda for the next generation of astronomical catalogs (LSST, Gaia DR4, Euclid, CMB-S4, DESI, and LIGO O4), grounded in the fluid mechanics of the vacuum as described by my Referential Relativity Theory (RRT). Recognizing that modern data processing pipelines often absorb real directional variances and reclassify them as artificial noise to preserve the premise of a perfectly isotropic universe, I formulate my predictions as "Dual Scenarios". For each observatory, I quantitatively establish: (A) the raw geometric and kinematic signatures expected from the causal anisotropic gradient, and (B) the specific algorithmic artifacts and physical conservation violations that will inevitably emerge if the academic community forces these new data to fit conventional dark matter models.

1 Phenomenological Foundation and Algorithmic Bias

My theory replaces the geometric singularity of the Big Bang with a Spontaneous Symmetry Breaking (SSB) that grants time an active, directional tensor structure ($T_{\mu\nu}$). The interaction between this causal flow and matter is governed by the Baryonic Neutrality Principle (BNP). This principle dictates that in high-density regions (such as our Solar System), temporal coupling saturates, perfectly recovering Einstein's local General Relativity. However, in the deep cosmic vacuum (Phase 3), spacetime acts as a viscous fluid governed by the universal impedance $\beta = 0.028006$.

The Variance Absorption Mechanism: The standard cosmological model habitually treats structural anisotropies as "noise." Calibration algorithms operate under the assumption that the universe must be isotropic. Mathematically, they minimize residuals by forcing the real anisotropic term (Δ_{aniso}) to zero. Similarly, early engineering notes from the LSST (Data Preview 1) already confess to systemic calibration flaws, such as over-scaling Right Ascension errors by the cosine of declination and failing aperture corrections for non-point structures.

- **Scenario A (RAW Data):** Unbiased photometry will demonstrate that the universe possesses a pure Temporal Anisotropy Gradient governed by the Law of Linear Causal Coupling ($D = D_0 \cdot z$). I have already confirmed this signature at an unprecedented 30.36σ significance using the SDSS DR16Q quasar catalog. It is imperative to note that this formidable statistical mark was extracted purely from empirical data through a strict Blind Audit Protocol, which deliberately shuffled phase residuals in Monte Carlo simulations to purge any prior theoretical extrapolation or spatial bias.

- **Scenario B (ΛCDM Manipulation):** As raw data passes through standard pipelines, the real causal signal will be "hijacked" and forced to artificially reappear as an anomalous excess in intrinsic error dispersion (σ_{int}) and color parameters. I predict that these catalogs will report an inexplicable excess of noise directionally aligned with the Cortez Axis (Cosmic Precession).

2 Fluid Dynamics in Stellar Streams (Gaia DR4 and LSST)

In the current academic paradigm, morphological "gaps" in stellar streams are phenomenologically attributed to random collisions with invisible dark matter sub-halos. RRT shifts this phenomenon from the stochastic to the deterministic domain. When a stellar stream plunges toward the galactic pericenter, its local acceleration drops below the Cortez Limit ($a_0 = 1.2001 \times 10^{-10}\,\text{m/s}^2$), and the system undergoes **Viscous Shear** due to the hydrodynamic friction of the $T_{\mu\nu}$ field.

Blind Falsification Test (CFT): The LSST and Gaia DR4 must map structural ruptures dictated by fluid mechanics in the exact following kilometric windows:

- Phlegethon: 12.5 kpc – 16.2 kpc
- Fjörm: 10.0 kpc – 14.3 kpc
- Slidr: 16.0 kpc – 19.8 kpc
- Sylgr: 13.0 kpc – 16.7 kpc
- Ylgr: 11.0 kpc – 14.9 kpc

The Dual Prediction:

- **Scenario A (Pure Astrometry):** Morphological gaps will occur exclusively at these *a priori* calculated coordinates. Furthermore, the 6D velocity dispersion at the edges of these gaps will reveal a strictly **laminar flow**, corresponding to a gradual tearing of the baryonic structure by spatial fluid tension.
- **Scenario B (N-Body Forcing):** When researchers attempt to explain these gaps by artificially inserting $10^5 M_\odot$ dark halos into their simulations, their standard models will generate a stochastic, thermal "spray" of stars. I predict this will violate the angular momentum conservation of the actual stream photographed by the telescopes, a principle my fluid dynamics framework preserves intrinsically.

3 Extreme Cosmological Precision Tests (Euclid, CMB-S4, and DESI)

To definitively dismiss the dark sector hypothesis, I offer the following mathematical predictions for the next generation of catalogs:

3.1 The Euclid and Roman Telescopes (Cosmic Shear)

The mapping of weak gravitational lensing by the Euclid satellite will not find the uniform web of dark matter expected by the consensus.

- **Scenario A:** The vector field $T_\mu(z)$ will induce a unique rotational signature (B-mode) in the shear maps. I establish that the cosmic shear will rigidly obey the equation $\gamma(\theta, z) = \gamma_{\text{ISO}}(\theta)\left[1 + A_T(z)\cos\big(2(\phi - \phi_T(z))\big)\right]$, with an exact projected amplitude of $A_T(z = 0.5) = 0.074 \pm 0.008$.

- **Scenario B:** Standard pipelines will initially try to classify this directional shear as galactic dust or telescope mirror aberration, forcing a local NFW profile until the mathematical discrepancy collapses the model.

3.2 Cosmic Microwave Background (CMB-S4) and Gaia Precession

I have demonstrated that the apparent "Axis of Evil" in the CMB is not a statistical anomaly, but the signature of the causal vector's orientation at recombination ($z = 1089$) with 98.36% precision.

- **Scenario A:** The CMB-S4 observatory will confirm, in the low multipoles ($\ell < 20$), an angular gradient of $h_* = 0.015 \pm 0.003$. For Gaia DR5/VLBI, I predict the real-time detection of Cosmic Rotation (Causal Axis Precession) at a rate of $d\phi_T/dt = (0.12 \pm 0.03)$ degrees per century.

3.3 Large-Scale Structures and Voids (DESI Catalog)

- **Scenario A (Void Cosmology Test - CFT-2):** Due to the Baryonic Neutrality Principle, the deep vacuum experiences less matter shielding. I predict that measuring cosmic expansion (H_0) restricted strictly to the interior of large cosmic voids will systematically yield a **higher H_0 value** ($K_{\text{eff}} \to 1$) than measurements taken within dense superclusters.
- **Scenario B:** Since ΛCDM cannot process topological variations in expansion, traditional measurements will report this variation as inexplicable parametric tensions in structural evolution ($f\sigma_8$).

4 The Causal Fatigue Test: Gravitational Wave Signatures (LIGO O4 / LISA)

My framework predicts that gravity, like light, suffers phase impedance from the viscous vacuum governed by the *Cortez Refractive Index*, $\eta_C(z) = 1 + \beta \cdot \ln(1 + z)$.

The Dual Prediction:

- **Scenario A (Clean Signal):** I establish the *Causal Fatigue Test*. Coalescence events (black holes or neutron stars) aligning with the Cortez Axis (Right Ascension $\approx 168°$) will register extreme causal energy dissipation, inducing a catastrophic divergence of up to **23%** between the inferred gravitational luminosity distance (D_L^{GW}) and the corresponding electromagnetic distance (D_L^{EM}).
- **Scenario B (ΛCDM Pipeline):** The LIGO collaboration frequently embeds assumed metric redshifts into their waveforms to calibrate distances. When algorithms mathematically force the signal to hide this distance "fatigue" by artificially lowering the frequency, it will result in an unreal inflation of the Chirp Mass. I predict the standard model will report a fantastical population of impossibly massive black holes mysteriously clustered at a single pole of the sky.

5 Conclusion: The End of Curve-Fitting

The era of adding infinitely malleable "dark parameters" to fix foundational geometric errors is coming to a close. The universe does not emerge from an infinite geometric singularity, but from a Spontaneous Symmetry Breaking that grants duration a vectorial tensor force ($T_{\mu\nu}$).

With the formalization of the core theory in early 2026, the Referential Relativity Theory has reached the ceiling of observational saturation. By unifying the cosmos under the constant β, I have reduced the galactic dynamic error to 1.33% (SPARC Catalog), achieved Theoretical Refractive Convergence in optical lensing at 99.89% (CASTLES Catalog), and secured a statistically definitive mark of 30.36σ for cosmic anisotropy.

Any future attempts to insert spherical, oblate, or triaxial dark matter halos over the raw data from Gaia DR4 and LSST will inevitably collapse under the laminar flow of the spatial macro-fluid. Cosmology must now abandon the geometry of shadows and firmly embrace the referential dynamics of real causal flows.